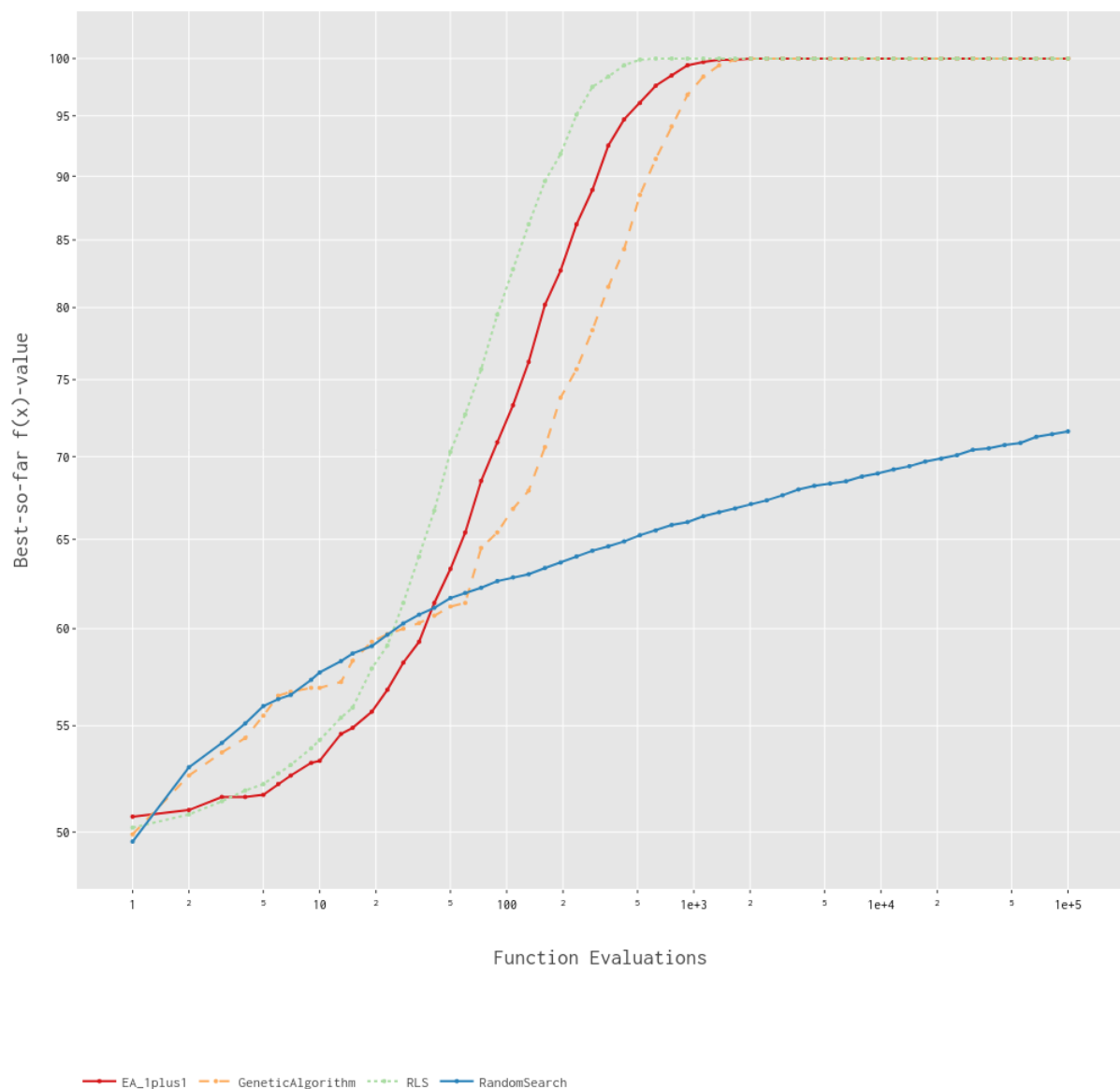
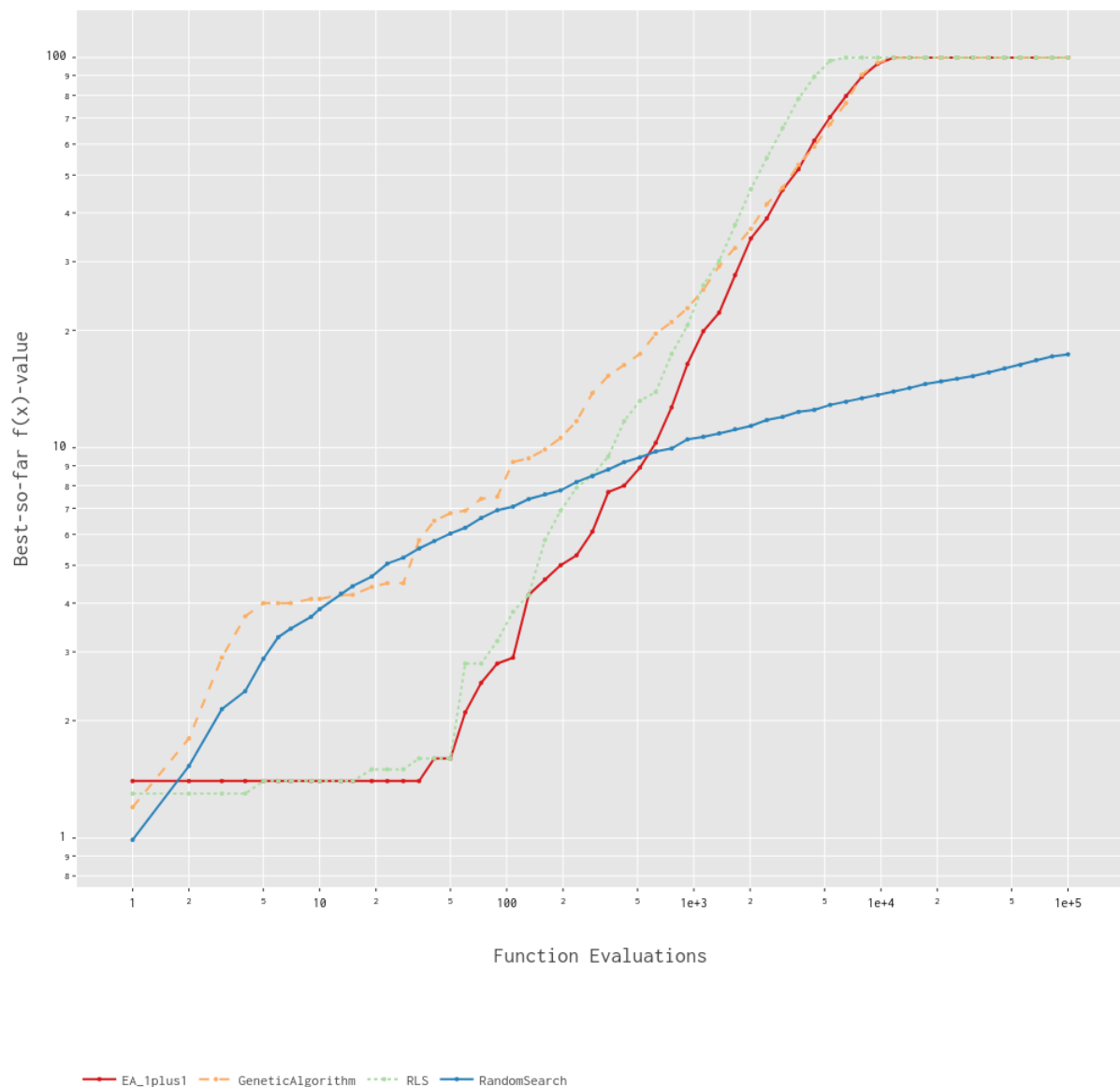


Random Search, RLS and EA (1+1), and Genetic Algorithm Plots with Summaries and Interesting Observations



Problem: F1 (OneMax)

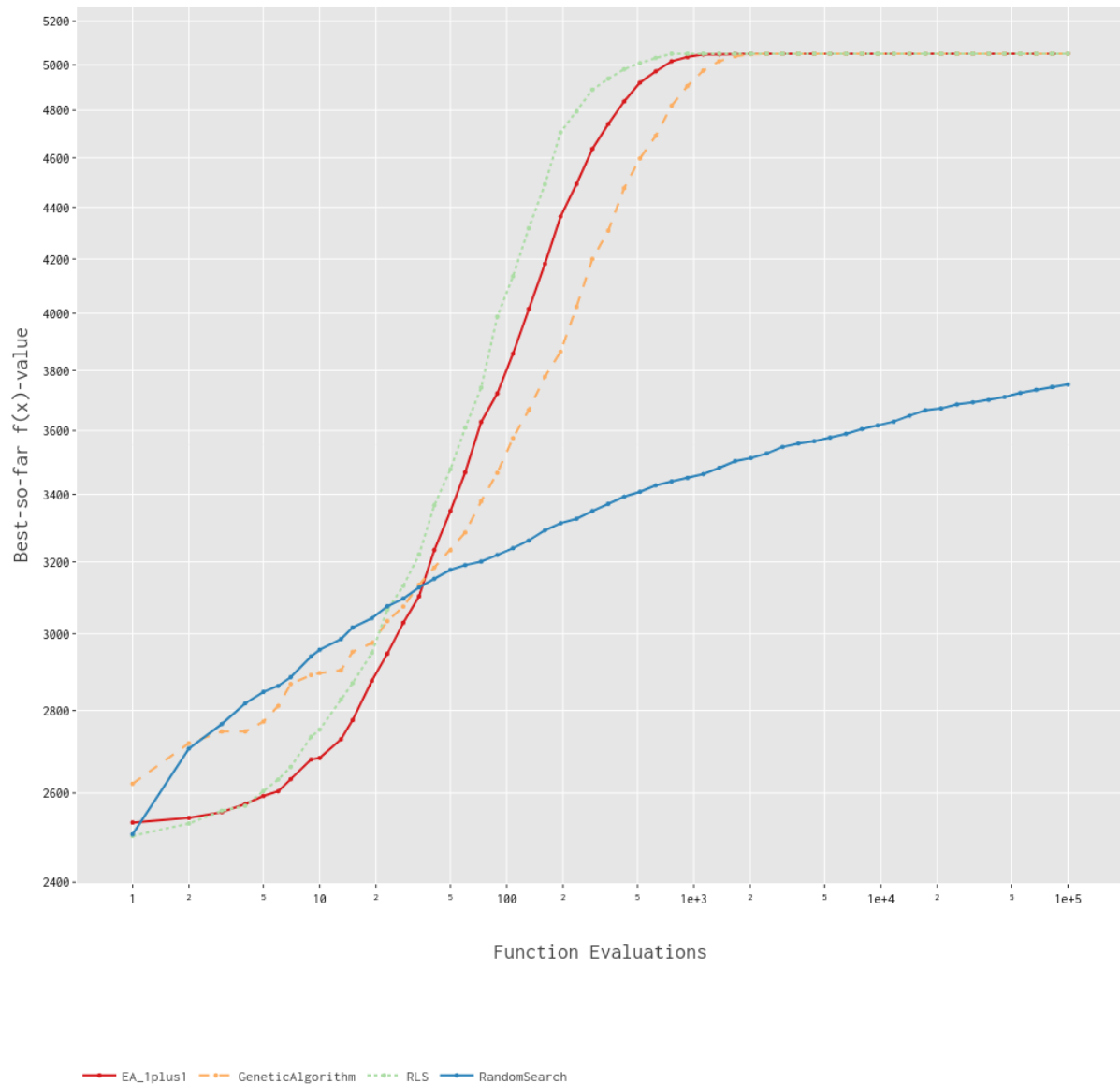
All algorithms show improvements in the number of ones in the bitstring as the number of function evaluations increases. RLS converges to the optimum value of 100 in around 500 evaluations while EA (1+1) and the Genetic Algorithm (GA) reach it in around 1000 and 1200 evaluations, respectively. Random Search failed to converge to the optimum and stopped at around 72. RLS was twice as fast as EA (1+1) and the GA, this is expected since OneMax rewards every single-bit improvement and RLS flips one bit per iteration. Both RLS and EA (1+1) show consistent performance when converging while Random Search shows steady but poor performance with little variation across runs. The GA performs competitively overall, finding the optimum through less stable, gradual growth.



Problem: F2 (LeadingOnes)

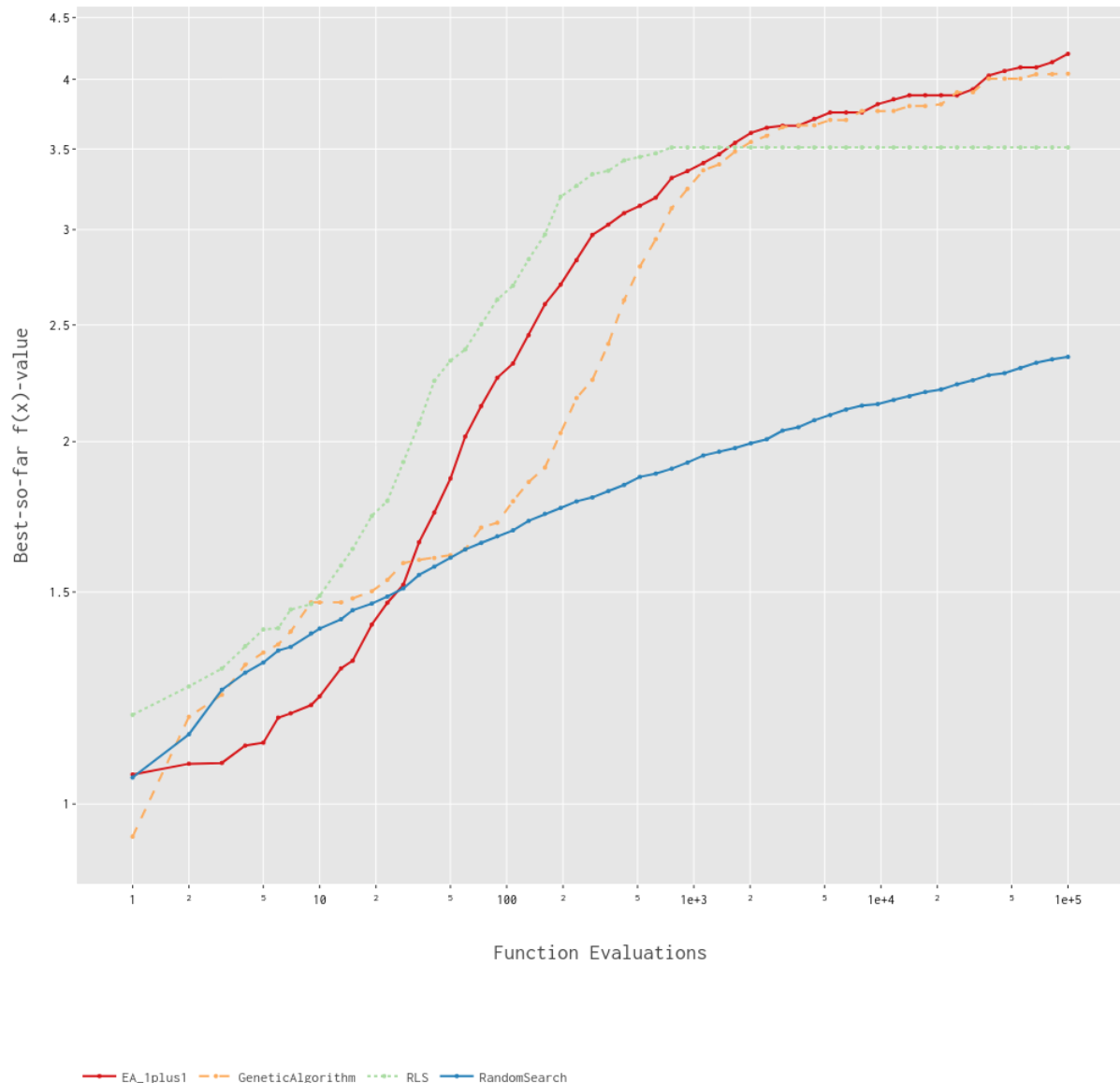
RLS converged to the optimum around 5000 evaluations, while the EA (1+1) reached it at around 10,000 evaluations. The GA also achieved the optimum at 10000 with a faster start but less stable growth. Random Search once again did not converge and ended at around 18.

LeadingOnes is more challenging than OneMax for all algorithms, as improvements depend on finding the next correct bit *in sequence*, causing progress to slow significantly once a wrong bit is encountered. Furthermore, RLS the EA (1+1) and the GA required roughly ten times more evaluations to reach the optimum compared to F1 which is consistent with the previous claim. RLS once again performed better than EA (1+1) and the GA by a factor of two, due to its single bit mutation strategy ensuring steady progress. The GA showed more unstable growth again likely due to the crossover occasionally breaking the leading prefixes before rediscovering them. What is most surprising here is the rate that the GA grows in early generations as it grows rapidly in comparison to the other algorithms. Although this is likely due to the many initial children which the GA utilises.



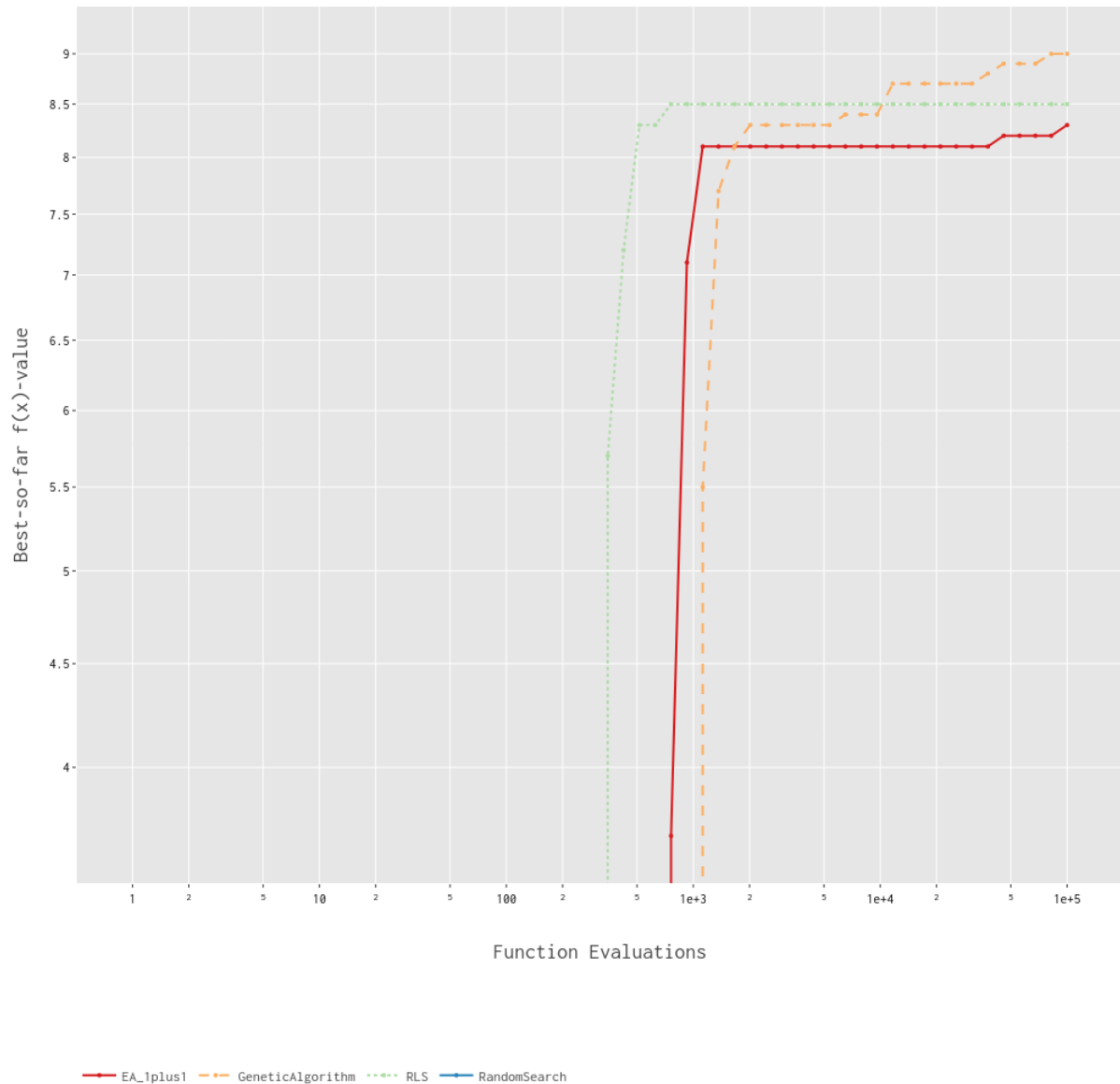
Problem: F3 (Linear Function with Harmonic Weights)

RLS reached the optimum of 5050 around 500 evaluations, EA (1+1) reached it around 1000 evaluations while GA followed closely behind reaching the optimum at 1200 evaluations. Random search once again did not reach the optimum, plateauing at around 3750 fitness. The performance pattern closely mirrors that of the OneMax problem, with nearly identical convergence speeds and algorithm rankings. This shows that the difference in bit weights does not affect the convergence speeds of these algorithms since they do not evaluate bit weights and treat them all equally.



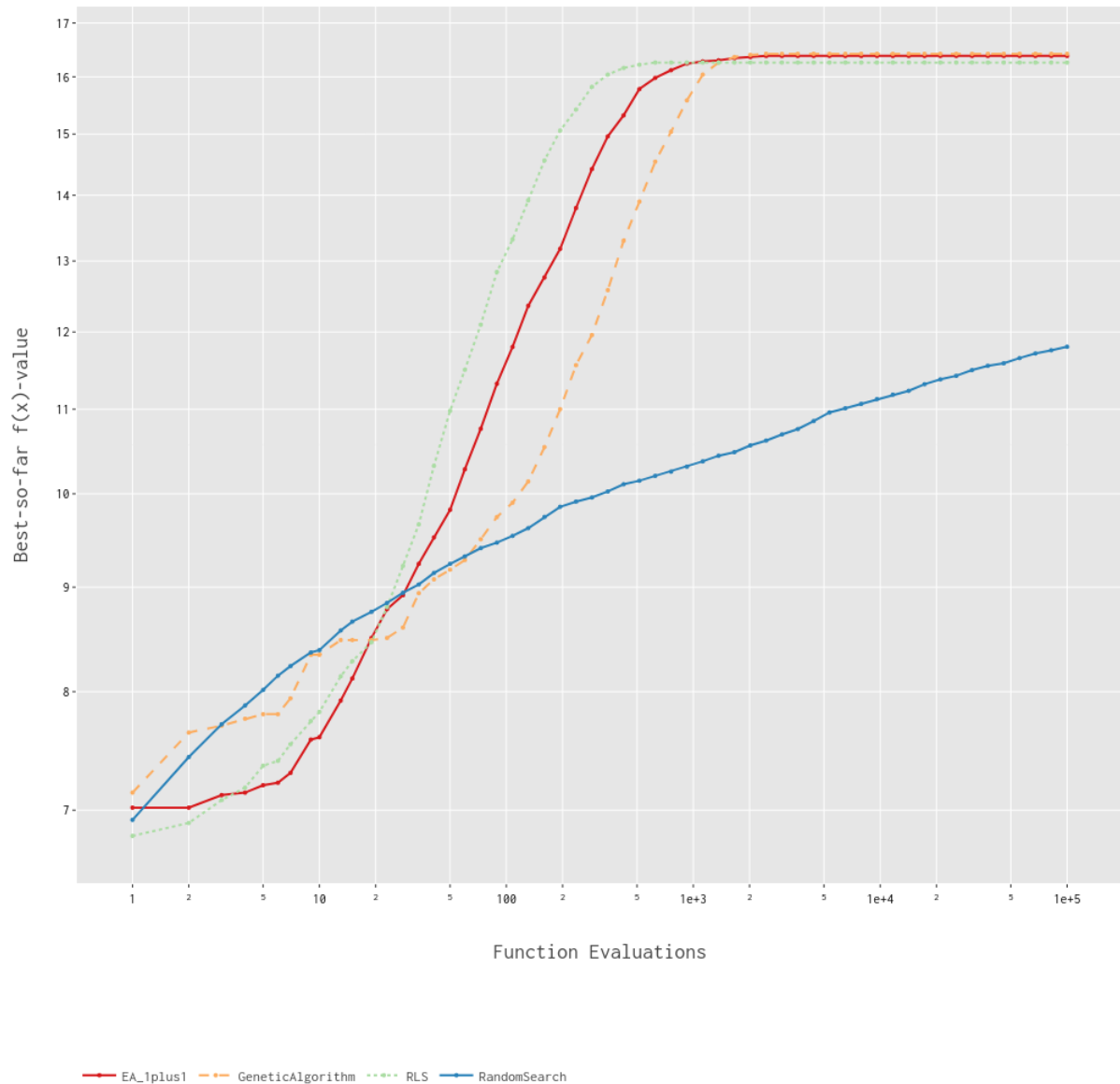
Problem: F18 (LABS)

None of the algorithms achieved the theoretical optimum. EA (1+1) achieved the best result at around 4.2 showing continuing improvement throughout. The Genetic Algorithm (GA) followed closely, showing similar upward progress and eventually converging slightly below the (1+1) EA at around 4.0. RLS plateaued at around 3.5 after 1000 evaluations while Random Search reached only around 2.4. This is the first problem where EA (1+1) outperformed RLS, highlighting the importance of broader exploration in more complex problems LABS is fundamentally more difficult than the linear functions due to its rugged fitness landscape and multiple local optima. RLS plateaued at 3.5 showing its limitations to escape once it's been trapped. RLS's early plateau demonstrates its limitation in escaping local traps, while both the (1+1) EA and GA benefited from their probabilistic search mechanisms and population diversity. High variance across runs suggests that initial conditions and random seed effects significantly influenced overall performance.



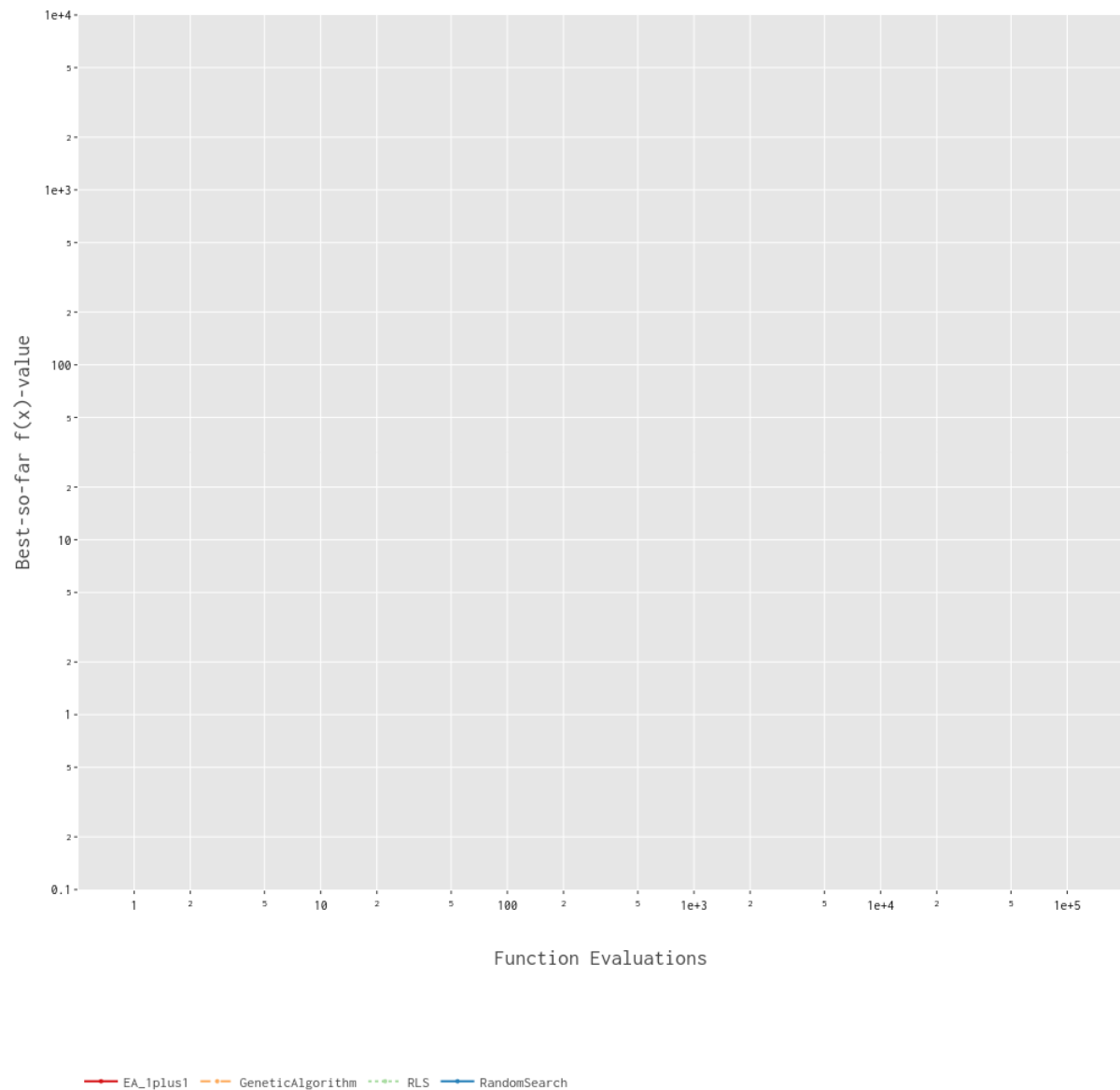
Problem: F23 (N-Queens)

RLS and EA (1+1) plateau rapidly around ~500 and ~1000 evaluations respectively. The Genetic Algorithm (GA) achieved similar performance, plateauing slightly later briefly at a comparable fitness before achieving the highest final fitness of 9 after around 15,000 evaluations. Random Search is not present in this graph as it plateaus much earlier than the other three algorithms and failed to find high-quality solutions within the evaluation limit. The rapid convergence of all effective algorithms suggests that once a near-correct sequence was discovered, substantial improvements followed almost instantly.



Problem: F24 (Concatenated Trap)

Both RLS, EA (1+1) and GA plateaued near the optimum value of 20, achieving just above ~16 in 500, 1000, and 1200 evaluations respectively. In contrast, Random Search maxed out just shy of 12 when it ran out of evaluations. The convergence behaviour closely resembles that seen in the OneMax and Linear Function with Harmonic Weights problems, indicating that local search remains highly effective for this problem type.



Problem: F25 (NKL)

Random Search, RLS, EA (1+1), and the Genetic Algorithm all failed to show any progress in this problem. This suggests that all the algorithms failed to find any meaningful improvements thus this problem is too difficult for them.